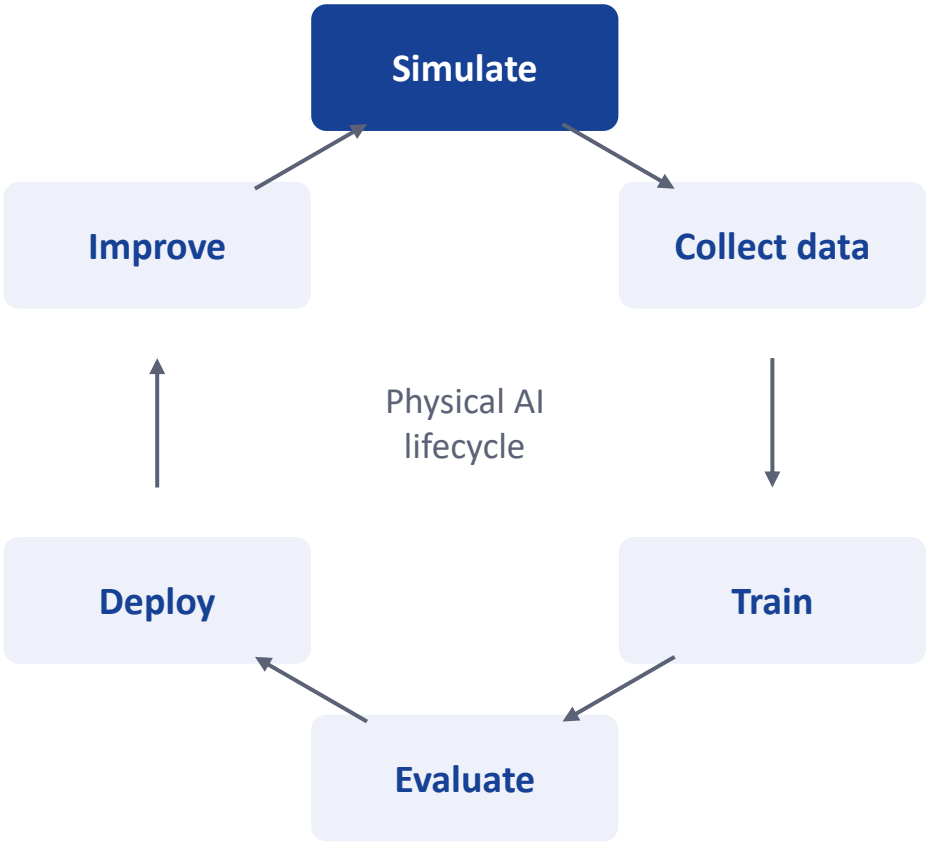
NANYANG
THE INNOVATIVE POLYTECHNIC

Physical AI: How a Robot Learns

Data Acquisition & Policy Learning · Lesson 1 (Lecture)



Our course map: one cycle, walked three times



- **A** Foundations L1–L2
- **B** Collecting robot data L3–L4
- **C** RL in simulation L5–L6
- **D** RL: sim → real arm L7–L8
- **E** Imitation learning L9–L10
- **F** Evaluate & deploy IL L11–L12
- **G** Vision-language-action L13–L14

Quick poll

What do you think is hard about getting a robot to do something useful?

Answer anonymously in the poll. One or two words is fine.

- **seeing?**
- **grasping?**
- **data?**
- **safety?**
- **cost?**

Learning outcomes

- By the end of this lesson you should be able to:
  - explain how physical AI differs from AI that runs on a screen
  - describe the six stages of the physical AI development lifecycle
  - name the observation and action of the SO-101 at each control step
  - compare reinforcement learning, imitation learning and VLA by where their training data comes from

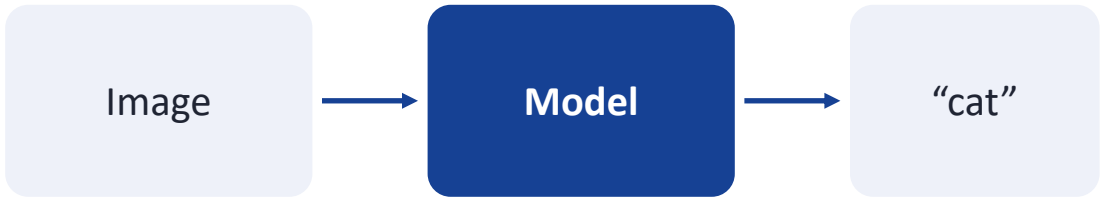
Part 1

From screen AI to physical AI

Why a robot is not just a classifier with a body

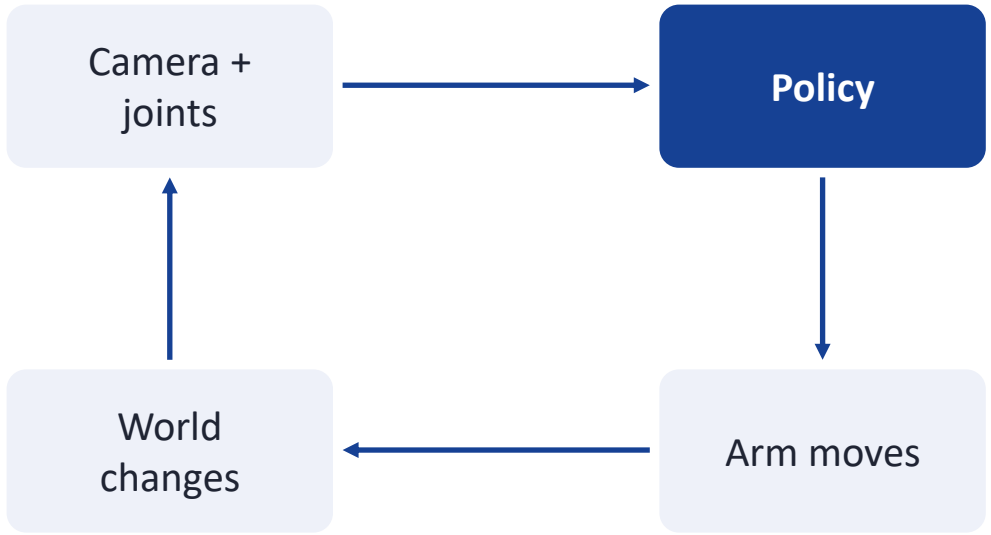
Screen AI answers once; a robot decides 30 times a second

Screen AI (e.g. image classifier)



One input, one answer.
The answer does not change the next image.
A mistake is just a wrong label.

Physical AI (robot policy)



Every action changes the next input.
A mistake moves the robot somewhere new.

The control loop: perceive → decide → act, every 33 ms



…and repeat. At 30 Hz there are 30 decisions per second: one every 33 milliseconds.

The policy is the only part we train. Everything else (the sensors and motors) stays the same for every method in this course.

Why physical AI is hard

Errors compound

2 mm off at the start of a grasp can become 2 cm off at the end. Each mistake feeds the next decision.

Data is not independent

Each frame follows from the last. Shuffling frames like images hides how the robot actually behaves.

Mistakes cost money

A bad action can hit the table, drop the object or strain a motor. Testing is never free.

Data is slow

One real demonstration takes about a minute. An image dataset can be scraped; robot data must be performed.

Simulation makes trials cheap, at a price

What simulation gives us

- Thousands of robots trying at once on one GPU
- Perfect labels for free (exact positions, contacts)
- No broken hardware, no safety risk
- Faster than real time

What it costs: the reality gap

Dynamics
masses, friction, how the motors respond

Appearance
colours, textures, lighting

Sensing
noise, offsets, delays

Anything the simulator leaves out, the policy never learns about.

Part 2

Robot basics with the SO-101

Joints, spaces, servos and one control step

Meet the SO-101

PHOTO / VIDEO
Labelled photo of your lab's SO-101 (leader and follower)

1. **shoulder_pan** rotates the whole arm left/right
2. **shoulder_lift** raises/lowers the upper arm
3. **elbow_flex** bends the elbow
4. **wrist_flex** tilts the wrist up/down
5. **wrist_roll** rotates the wrist
6. **gripper** opens and closes the jaws

Open-source design by TheRobotStudio with Hugging Face

Joints, degrees of freedom and workspace

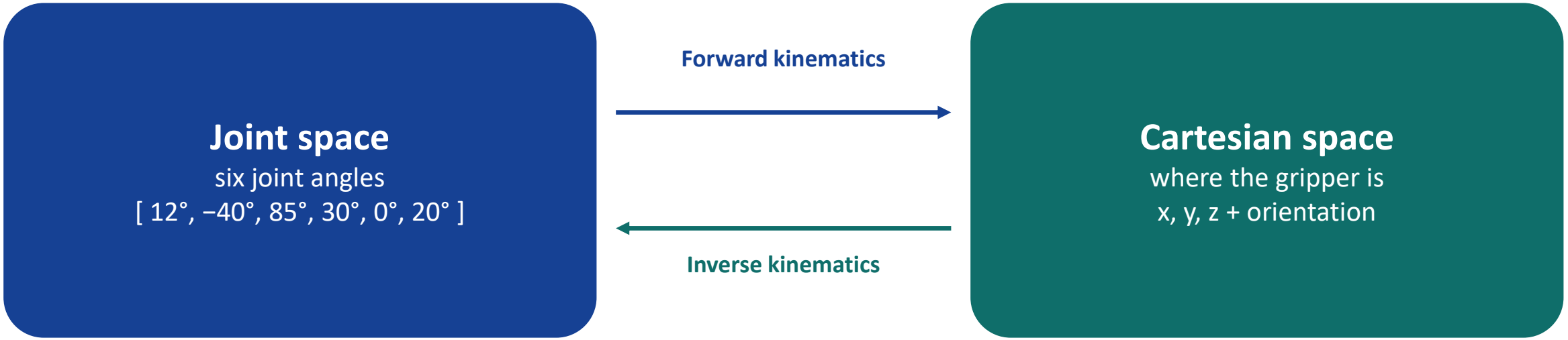
Joint
A place where two parts of the arm connect and can move.

Degree of freedom (DOF)
An independent way the robot can move. The SO-101 has 5 arm DOF plus a gripper.

Workspace
All the places the gripper can reach. Outside it, no policy can succeed.

PHOTO / VIDEO
Short clip: moving one joint at a time

Two ways to describe the same pose



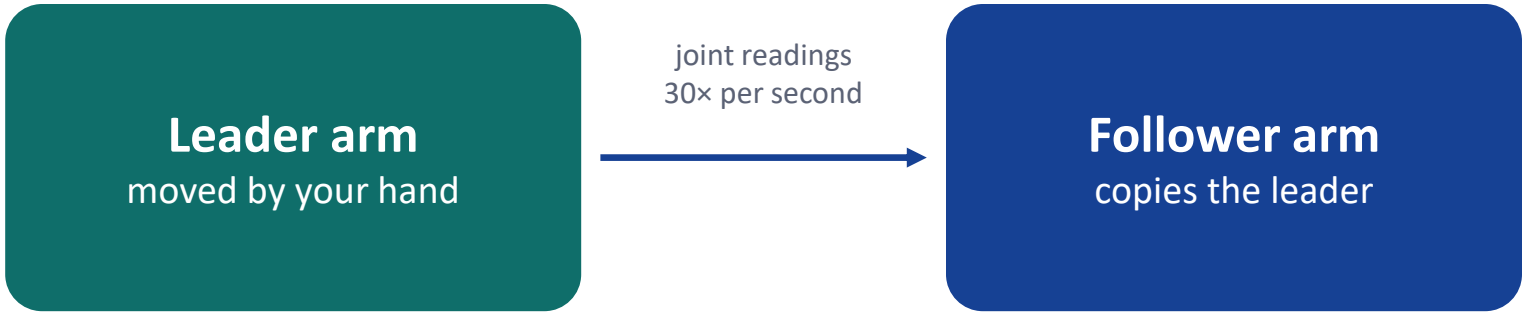
Forward: “Given how my arm is bent, where is my hand?” (one answer)
Inverse: “How should I bend my arm to reach the cup?” (maybe many answers, maybe none)

Our servos take positions, not forces

We CAN command	We CANNOT command
A target position for each joint (the servo's own controller drives towards it)	A force or torque “Grip harder” or “push gently”

Six Feetech STS3215 serial-bus servos. Grip force comes only indirectly, by telling the gripper to close further than the object allows. This limit returns in Lessons 7 and 9.

Leader, follower and cameras



Front camera
sees the whole scene

Wrist camera
sees the grasp up close

Recorded every frame: leader position (the action), follower position (the state), both camera images.
This is how most of the data in this course will be collected.

Worked example: one control step

Observation (input)

6 joint positions
[2.1, −38.5, 81.0, 27.4, −1.2, 18.0]
+ front image + wrist image

policy →

Action (output)

6 target joint positions
[2.4, −37.9, 80.2, 27.9, −1.2, 12.5]

180
joint numbers per second

60
images per second

600
frames in a 20 s demo

1,200
images in a 20 s demo

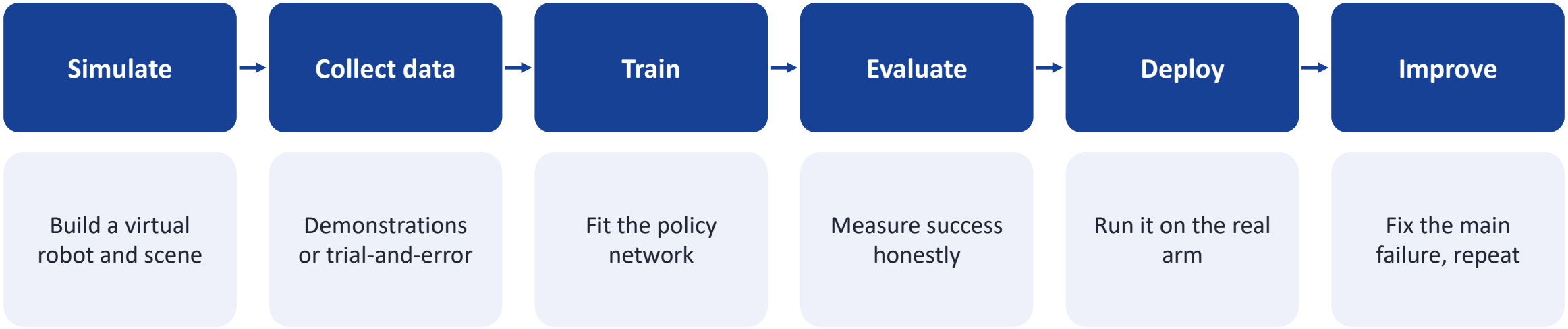
Example values are illustrative; replace with numbers captured from your own arm.

Part 3

The lifecycle and three ways to train an arm

Where does the training data come from?

The physical AI development lifecycle



Simulation helps most at the start. Evaluation and improvement are where most real projects spend their time.

Three ways to train an arm

	Reinforcement learning	Imitation learning	Vision-language-action
Where data comes from	Robot tries, gets a score	Human demonstrates	Demos + a sentence
What you provide	You write the reward	You record the dataset	Fine-tune a big pretrained model
Where it trains	Mostly in simulation	Real arm, no reward	One model, many tasks
Typical failure	Fails by reward hacking	Fails off its training data	Can pick the wrong task

The key question: does the data come from the robot trying, or from a person showing?

Course roadmap: 14 weeks

Foundations		Data		RL in sim		RL → arm		Imitation		Evaluate		VLA	
Wk 1 **L1** Lec	Wk 2 **L2** Lab	Wk 3 **L3** Lec	Wk 4 **L4** Lab	Wk 5 **L5** Lec	Wk 6 **L6** Lab	Wk 7 **L7** Lec	Wk 8 **L8** Lab	Wk 9 **L9** Lec	Wk 10 **L10** Lab	Wk 11 **L11** Lec	Wk 12 **L12** Lab	Wk 13 **L13** Lec	Wk 14 **L14** Lab

Test 1
after Week 8 · Lessons 1–8

Test 2
after Week 14 · Lessons 9–14

Assignment
pick-and-place project · demo after Week 14

Lecture weeks explain the idea; lab weeks make it run in Isaac Sim and on the SO-101. Each lesson starts with a short check on the last one.

Activity: Spot the stage

In pairs, label each card with a lifecycle stage and the method it belongs to (RL, IL or VLA).

“A student hand-guides the leader arm fifty times.”
Answer: Collect data · IL

“4,096 simulated arms try to reach a target.”
Answer: Train · RL

“The policy succeeds 14 times out of 20 on the real arm.”
Answer: Evaluate · any

“Each demo is saved with ‘pick the blue cube’.”
Answer: Collect data · VLA

10 minutes · 8 cards per pair · answers shown on screen afterwards

Exit ticket

- Answer on paper or in the form before you leave:
  - What does the SO-101 policy receive as its observation each step?
  - What does it output as its action?
  - Put the six lifecycle stages in order.
  - Where does reinforcement learning get its training data?
  - Where does imitation learning get its training data?

Next week: L2 lab · Meet the tools

Sim station · Isaac Sim

Tour the simulator
Load the SO-101 and move each joint
Watch a trained Reach policy

Arm station · SO-101

Calibrate leader and follower
Teleoperate with live camera view
Fill in the joint table

Safety first: clear workspace, hands out of the follower's path, cut the power (not just USB) to stop, never leave a powered arm alone.

References and further reading

- Capuano et al. (2025). Robot Learning: A Tutorial. https://arxiv.org/abs/2510.12403
- Hugging Face LeRobot documentation: SO-101. https://huggingface.co/docs/lerobot/so101
- NVIDIA. Train an SO-101 Robot From Sim-to-Real With NVIDIA Isaac. https://docs.nvidia.com/learning/physical-ai/sim-to-real-so-101/latest/index.html
- Zhao et al. (2023). Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware (ACT). https://arxiv.org/abs/2304.13705
- Rudin et al. (2021). Learning to Walk in Minutes Using Massively Parallel Deep RL. https://arxiv.org/abs/2109.11978
- Lynch & Park. Modern Robotics (free book and videos). https://hades.mech.northwestern.edu/index.php/Modern_Robotics